

Entrega Tema 1

Métodos Matemáticos Para la Física

Problema 1 (4 puntos)

Dibuja el campo de pendientes para las siguientes ecuaciones diferenciales de primer orden. Explica el comportamiento cuando x crece y dibuja una solución (curva integral) haciendo uso del campo de pendientes obtenido.

a) $y' = e^{x-y}$

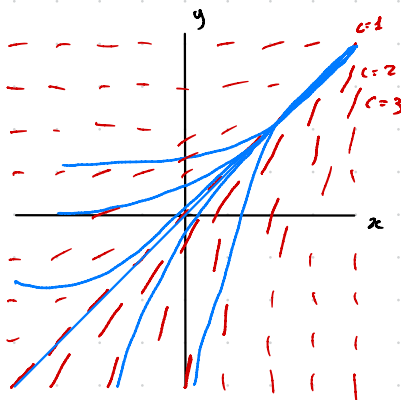
b) $y' = x^2$

c) $y' = \frac{1}{x}$

d) $y' = \cos y$

a) $y' = e^{x-y}$ $x - y = \ln(c) \rightarrow y = x - \ln(c)$

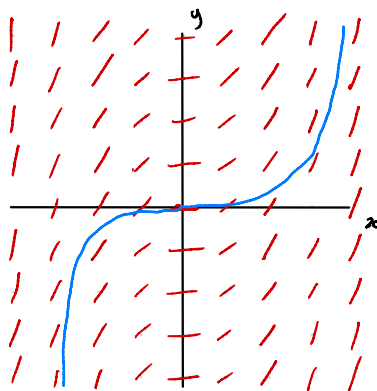
c	$F(x)$
1	$y = x$
2	$y = x - \ln(2)$



Cuando x crece la función se acerca a $y = x$

b) $y' = x^2 \rightarrow c = x^3$

c	$F(x)$
0	$x^2 = 0 \rightarrow x = 0$
1	$x^2 = 1 \rightarrow x = \pm 1$
4	$x^2 = 4 \rightarrow x = \pm 2$
9	$x^2 = 9 \rightarrow x = \pm 3$
16	$x^2 = 16 \rightarrow x = \pm 4$

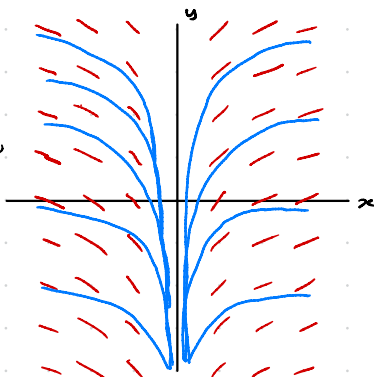


Cuando x crece, y crece cúbicamente
 $y = x^3$

c) $y' = \frac{1}{x} \rightarrow c = \frac{1}{x} \rightarrow x = \frac{1}{c}$

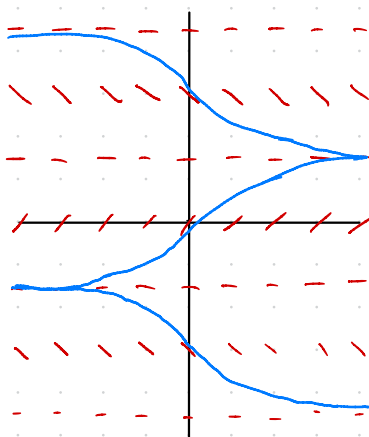
c	F(x)
1	$x = 1$
-1	$x = -1$
$-\frac{1}{2}$	$x = -2$
$\frac{1}{2}$	$x = 2$
$\frac{1}{3}$	$x = 3$
$-\frac{1}{3}$	$x = -3$

Asintota en $x=0$,
cuando x crece
se acerca a $x=0$
Si $x < 0$ y se aleja
cuando $x > 0$



d) $y' = \cos y$

c	F(x)
0	$y = \pi/2, 3\pi/2, -\pi/2, -3\pi/2$
1	$y = 0, 2\pi, -2\pi$
-1	$y = \pi, 3\pi, -3\pi$



Si $y \in (-\pi/2, \pi/2)$,
 x es creciente y se
acerca a $y = \pi/2$

Si $y \in (\pi/2, 3\pi/2)$
decreciente y se acerca
a x a $y = \pi/2$

Este patrón se repite
cada $y = n\pi$ radianes
se acerca a $y = \frac{n\pi}{2}$

Problema 2 (4 puntos)

Resuelve las siguientes ecuaciones diferencial separables:

a) $x^2 y' = 1 - x^2 + y^2 - x^2 y^2, y(1) = 0$

b) $\frac{dx}{dt} = x \sin(t), \text{ for } x(0) = 1$

c) $\frac{dx}{dt} = x \sin(t), \text{ for } x(0) = 1$

d) $y' = 2xy$

a) $x^2 \frac{dy}{dx} = 1 - x^2 + y^2 - x^2 y^2$

$$\frac{dy}{dx} = \frac{1}{x^2} - 1 + \frac{y^2}{x^2} - y^2 = \left(\frac{1}{x^2} - 1\right) + y^2 \left(\frac{1}{x^2} - 1\right) = (1+y^2) \left(\frac{1}{x^2} - 1\right)$$

$$\frac{1}{1+y^2} \frac{dy}{dx} = \frac{1}{x^2} - 1 \rightarrow \frac{1}{1+y^2} dy = \left(\frac{1}{x^2} - 1\right) dx \rightarrow \int \frac{1}{1+y^2} dy = \int \left(\frac{1}{x^2} - 1\right) dx$$

$$\arctan(y) = -\frac{1}{x} - x + C \rightarrow y = \tan\left(-\frac{1}{x} - x + C\right)$$

$$y(1) = 0 \rightarrow 0 = \tan\left(-1 - 1 + C\right) \rightarrow 0 = \tan(C - 2) \rightarrow C = 2$$

$$y(x) = \tan\left(2 - \frac{1}{x} - x\right)$$

c) b) $\frac{dx}{dt} = x \sin(t) \rightarrow \frac{1}{x} \frac{dx}{dt} = \sin(t) \rightarrow \int \frac{1}{x} dx = \int \sin(t) dt$

$$\ln(x) = -\cos(t) + C \rightarrow x = e^{-\cos(t) + C} = A e^{-\cos(t)} \quad x(0) = 1$$

$$1 = A \cdot e^{-\cos(0)} \Rightarrow 1 = \frac{A}{e} \rightarrow A = e \rightarrow x(t) = e \cdot e^{-\cos(t)}$$

$$x(t) = e^{1 - \cos(t)}$$

d) $y' = 2xy \rightarrow \frac{dy}{y} = 2x dx \rightarrow \int \frac{1}{y} dy = \int 2x dx \rightarrow \ln(y) = x^2 + C$

$$y = e^{x^2 + C} = A e^{x^2}$$

Problema 3 (2 puntos)

Se tiene una taza de café a 100°C en el tiempo $t=0$, y su temperatura se reduce a 70°C a $t=10$ minutos, y a 50°C para $t=20$ minutos. Calcular la temperatura ambiente.

$$\frac{dT}{dt} = -k(T - T_A) \rightarrow \frac{1}{T - T_A} \frac{dT}{dt} = -k \rightarrow \frac{1}{T - T_A} dT = -k dt$$

$$\int \frac{1}{T - T_A} dT = \int -k dt \rightarrow \ln|T - T_A| = -kt + C \rightarrow T - T_A = Ce^{-kt}$$

$$T(t) = T_A + Ce^{-kt}, \quad T(0) = 100, \quad T(10) = 70, \quad T(20) = 50$$

$$100 = T_A + Ce^{-0t} = T_A + C, \quad 70 = T_A + Ce^{-10k}, \quad 50 = T_A + Ce^{-20k}$$

$$C = 100 - T_A \rightarrow 70 = T_A + (100 - T_A)e^{-10k}, \quad 50 = T_A + Ce^{-20k}$$

$$\frac{70 - T_A}{50 - T_A} = \frac{(100 - T_A)Ce^{-10k}}{(100 - T_A)Ce^{-20k}} \rightarrow \frac{70 - T_A}{50 - T_A} = e^{10k} \rightarrow k = \frac{1}{10} \ln \left| \frac{70 - T_A}{50 - T_A} \right|$$

$$70 - T_A = (100 - T_A)e^{-10k} = (100 - T_A)e^{-\ln \left| \frac{70 - T_A}{50 - T_A} \right|} = (100 - T_A) \frac{50 - T_A}{70 - T_A}$$

$$(70 - T_A)^2 = (100 - T_A)(50 - T_A) \rightarrow T_A^2 - 140T_A + 4900 = T_A^2 - 150T_A + 5000$$
$$10T_A = 1000 \rightarrow T_A = 10^{\circ}\text{C}$$

Problema 4 (5 puntos)

Resuelve las siguientes ecuaciones diferenciales mediante el factor integrante:

- a) $y' + xy = x$
- b) $y' + 6y = e^x$
- c) $y' + 3x^2y = \sin(x)e^{-x^3}$, con $y(0) = 1$
- d) $y' + \cos(x)y = \cos(x)$
- e) $\frac{1}{x^2+1}y' + xy = 3$, con $y(0) = 0$.

a) $y' + xy = x \quad \begin{cases} P(x) = x \\ Q(x) = x \end{cases}$

Factor Integrante: $P(x) = e^{\int x dx} = e^{x^2/2}$

$$e^{x^2/2}y' + xe^{x^2/2}y = xe^{x^2/2} \rightarrow \frac{d(e^{x^2/2}y)}{dx} = xe^{x^2/2} \rightarrow \int \frac{d(e^{x^2/2}y)}{dx} = \int xe^{x^2/2} dx$$

$$e^{x^2/2}y = e^{x^2/2} + C \rightarrow y = 1 + Ce^{-x^2/2}$$

b) $y' + 6y = e^x \rightarrow \begin{cases} P(x) = 6 \\ Q(x) = e^x \end{cases}$

$$P(x) = e^{\int 6 dx} = e^{6x} \rightarrow e^{6x}y' + 6e^{6x}y = e^{6x} \cdot e^x = e^{7x}$$

$$\frac{d(e^{6x}y)}{dx} = e^{7x} \rightarrow e^{6x}y = \int e^{7x} dx \rightarrow e^{6x}y = \frac{1}{7}e^{7x} + C$$

$$y = 7 \cdot \frac{e^{7x}}{e^{6x}} + Ce^{-6x} = 7e^x + Ce^{-6x}$$

c) $y' + 3x^2y = \sin(x)e^{-x^3} \quad y(0) = 1 \quad P(x) = 3x^2 \quad Q(x) = \sin(x)e^{-x^3}$

$$P(x) = e^{\int 3x^2 dx} = e^{x^3} \rightarrow e^{x^3}y' + 3x^2e^{x^3}y = \sin(x)e^{-x^3} \cdot e^{x^3} = \sin(x)$$

$$\frac{d(e^{x^3}y)}{dx} = \sin(x) \rightarrow e^{x^3}y = \int \sin(x) dx \rightarrow e^{x^3}y = -\cos(x) + C$$

$$y = e^{-x^3}(-\cos(x) + C) \quad y(0) = 1 \rightarrow 1 = e^0(-\cos(0) + C) \rightarrow 1 = -1 + C$$
$$C = 2 \rightarrow y(x) = e^{-x^3}(-\cos(x) + 2)$$

$$d) y' + \cos(x)y = \cos(x) \quad P(x) = \cos(x) \quad Q(x) = \cos(x)$$

$$P(x) = e^{\int \cos(x) dx} = e^{\sin(x)} \rightarrow e^{\sin(x)} y' + \cos(x) e^{\sin(x)} y = \cos(x) e^{\sin(x)}$$

$$\frac{d(e^{\sin(x)} y)}{dx} = \cos(x) e^{\sin(x)} \rightarrow e^{\sin(x)} y = \int \cos(x) e^{\sin(x)} dx = e^{\sin(x)} + C$$

$$y = 1 + C e^{-\sin(x)}$$

$$e) \frac{1}{x^2+1} y' + x y = 3 \quad y(0) = 0$$

$$y' + x(x^2+1) = 3(x^2+1) \quad P(x) = x^3+x \quad Q(x) = 3x^2+3$$

$$P(x) = e^{\int x^3+x dx} = e^{\frac{x^4}{4} + \frac{x^2}{2}}$$

$$e^{\frac{x^4}{4} + \frac{x^2}{2}} y' + (x^3+x) e^{\frac{x^4}{4} + \frac{x^2}{2}} = (3x^2+3) e^{\frac{x^4}{4} + \frac{x^2}{2}}$$

$$e^{\frac{x^4}{4} + \frac{x^2}{2}} y = \int (3x^2+3) e^{\frac{x^4}{4} + \frac{x^2}{2}} dx \rightarrow \text{L'Integral de la droite nous donne sol}$$

$$y = \frac{\int_0^x 3(t^2+1) e^{\frac{t^4}{4} + \frac{t^2}{2}} dt}{e^{\frac{x^4}{4} + \frac{x^2}{2}}} \quad \begin{matrix} \nearrow \\ \text{+ } y_0 \end{matrix} \rightarrow 0 \rightarrow y_0 = 0$$

Problema 5 (5 puntos)

Resuelve las siguientes ecuaciones diferenciales mediante sustitución:

a) $xy' + y(x+1) + xy^5 = 0, \quad y(1) = 1.$

b) $x^2y' = y^2 + xy, \quad y(1) = 1.$

c) $y' = (x - y + 1)^2$

d) $y' + y(x^2 - 1) + xy^6 = 0, \text{ con } y(1) = 1$

e) $2yy' + 1 = y^2 + x, \text{ con } y(0) = 1$

a) $xy' + y(x+1) + xy^5 = 0 \quad y(1) = 1$

$$y' + \frac{(x+1)}{x} y + y^5 = 0 \rightarrow y' + \frac{(x+1)}{x} y = -y^5$$

Bernoulli: $\begin{cases} p(x) = \frac{x+1}{x} \\ q(x) = -\frac{1}{x} \end{cases} \quad v = y^{-4} \rightarrow y = v^{-1/4}$

$$\frac{dy}{dx} = -\frac{1}{4} v^{-5/4} \frac{dv}{dx}$$

$$-\frac{1}{4} v^{-5/4} v' + \frac{(x+1)}{x} v^{-1/4} = -v^{-5/4} \rightarrow v' - 4 \frac{(x+1)}{x} v = 4 \quad \begin{cases} p(x) = 4 \frac{(x+1)}{x} \\ q(x) = 4 \end{cases}$$

$$p(x) = e^{\int -4 \frac{(x+1)}{x} dx} = e^{-4 \left(1 + \frac{1}{x}\right)} = e^{-4x} \cdot x^{-4}$$

$$x^{-4} e^{-4x} v' + 4 x^{-4} e^{-4x} \frac{(x+1)}{x} v = 4 e^{-4x} x^{-4}$$

$$x^{-4} e^{-4x} v = \int 4 e^{4x} x^{-4} dx \rightarrow \text{Esta Integral no tiene solución}$$

$$v = e^{4x} x^4 \cdot \int_1^x 4 e^{-4t} t^{-4} dt + 1 \quad v = y^{-4}$$

$$y = e^{-x} \cdot x \cdot \left(\int_1^x 4 e^{-4t} t^{-4} dt + 1 \right)^{1/4}$$

$$b) x^2 y' = y^2 + x y$$

$$x^2 y' - x y = y^2 \rightarrow y' - \frac{y}{x} = \frac{y^2}{x^2} \quad \begin{cases} n=2 \\ P(x) = -1/x \\ Q(x) = 1/x^2 \end{cases}$$

$$v = y^{1-n} \rightarrow v = y^{-1} \rightarrow y = \frac{1}{v} \rightarrow \frac{dy}{dx} = \frac{dy}{dv} \frac{dv}{dx} = -\frac{1}{v^2} \frac{dv}{dx}$$

$$-\frac{1}{v^2} \frac{dv}{dx} - \frac{1}{x} \frac{1}{v} = \frac{1}{x^2} \cdot \frac{1}{v^2} \rightarrow \frac{dv}{dx} + \frac{1}{x} v = -\frac{1}{x^2} \quad \begin{cases} P(x) = 1/x \\ Q(x) = -1/x^2 \end{cases}$$

$$p(x) = e^{\int 1/x dx} = e^{\ln|x|} = x$$

$$x \frac{dv}{dx} + v = -\frac{1}{x} \rightarrow x v = \int -1/x = -\ln|x| + C \rightarrow v = \frac{-\ln|x| + C}{x}$$

$$y = v^{-1} \rightarrow y = \frac{x}{-\ln|x| + C} \quad y(1) = 1 \rightarrow -\ln|1| + C = 1 \rightarrow C = 1$$

$$y = \frac{x}{\ln|x| + 1}$$

$$c) y' = (x - y + 1)^2$$

$$v = x - y + 1 \rightarrow y = x - v + 1 \quad \frac{dy}{dx} = 1 - \frac{dv}{dx}$$

$$\frac{dy}{dx} = v^2 \rightarrow 1 - \frac{dv}{dx} = v^2 \rightarrow \frac{dv}{dx} = 1 - v^2$$

$$\frac{1}{1-v^2} \frac{dv}{dx} = 1 \rightarrow \int \frac{1}{1-v^2} dv = \int dx$$

$$\int \frac{1}{1-v^2} = \int \frac{1}{(1-v)(1+v)} dv$$

$$\frac{A}{(1-v)} + \frac{B}{(1+v)} = \frac{1}{(1-v)(1+v)}$$

$$A(1+v) + B(1-v) = 1$$

$$v = 1 \rightarrow 2A = 1 \rightarrow A = 1/2 \quad v = -1 \quad 2B = 1 \rightarrow B = 1/2$$

$$\int \frac{1}{2(1-v)} + \frac{1}{2(1+v)} dv = \int dx \rightarrow \frac{1}{2} \ln|1-v| + \frac{1}{2} \ln|1+v| = x + C$$

$$\frac{1}{2} \ln|(1-v)(1+v)| = x + C \rightarrow \ln|\sqrt{1-v^2}| = x + C \rightarrow \sqrt{1-v^2} = e^{x+C}$$

$$1-v^2 = (e^x e^C)^2 \rightarrow 1-v^2 = A^2 e^{2x} \rightarrow 1 - A^2 e^{2x} = v^2 \rightarrow v = \pm \sqrt{1 - A^2 e^{2x}}$$

$$x - y + 1 = \sqrt{1 - A^2 e^{2x}} \rightarrow y = 1 + x \mp \sqrt{1 - A^2 e^{2x}}$$

$$d) y' + (x^2 - 1)y + xy^6 = 0 \quad y(1) = 1$$

$$y' + (x^2 - 1)y = -xy^6 \rightarrow \text{Bernoulli} \begin{cases} n=6 \\ P(x) = x^2 - 1 \\ Q(x) = -x \end{cases}$$

$$v = y^{1-6} = y^{-5} \rightarrow y = v^{-1/5} \quad y' = \beta_x^0 + \beta_v \frac{dv}{dx} \rightarrow y' = -1/5 v^{-6/5} \frac{dv}{dx}$$

$$-1/5 v^{-6/5} \frac{dv}{dx} + (x^2 - 1)v^{-1/5} = -x v^{-6/5} \rightarrow -1/5 \frac{dv}{dx} + (x^2 + 1)v = -x$$

$$\frac{dv}{dx} - 5(x^2 - 1)v = 5x \quad \begin{cases} P(x) = -5(x^2 - 1) \\ Q(x) = 5x \end{cases} \quad P(x) = e^{\int -5(x^2 - 1)dx} = e^{-\frac{5x^3}{3} + 5x}$$

$$e^{-5(\frac{x^3}{3} - x)} \frac{dv}{dx} - 5(x^2 - 1)e^{-5(\frac{x^3}{3} - x)} = 5x e^{-5(\frac{x^3}{3} - x)} \rightarrow \frac{d(e^{-5(\frac{x^3}{3} - x)} v)}{dx} = 5x e^{-5(\frac{x^3}{3} - x)}$$

$$e^{-5(\frac{x^3}{3} - x)} v = \int 5x e^{-5(\frac{x^3}{3} - x)} dx \rightarrow \text{Esta Integral no tiene solución}$$

$$y = \left(e^{5(\frac{x^3}{3} - x)} \left(\int_1^x 5t e^{-5(\frac{t^3}{3} - t)} dt + 1 \right) \right)^{1/5}$$

$$e) 2yy' + 1 = y^2 + x \quad y(0) = 1$$

$$2yy' - y^2 = x - 1 \rightarrow y' - \frac{y}{2} = \frac{1}{2}(x - 1)y^{-1} \rightarrow \text{Bernoulli} \rightarrow \begin{cases} n = -1/2 \\ P(x) = -1/2 \\ Q(x) = 1/2(x - 1) \end{cases}$$

$$v = y^2 \rightarrow y = \sqrt{v} = v^{1/2} \quad \frac{dy}{dx} = 1/2 v^{-1/2} \frac{dv}{dx}$$

$$1/2 v^{-1/2} \frac{dv}{dx} - \frac{v^{1/2}}{2} = \frac{1}{2}(x - 1)v^{-1/2} \rightarrow \frac{dv}{dx} - v = (x - 1) \quad \begin{cases} P(x) = -1 \\ Q(x) = (x - 1) \end{cases}$$

$$P(x) = e^{\int P(x)dx} = e^{\int -1 dx} = e^{-x} \rightarrow e^{-x} \frac{dv}{dx} - e^{-x} v = (x - 1)e^{-x}$$

$$\frac{d(e^{-x} v)}{dx} = (x - 1)e^{-x} \rightarrow e^{-x} v = \int (x - 1)e^{-x} dx \rightarrow e^{-x} v = -x e^{-x} - 1 \rightarrow v = -x + (e^{-x})$$

$$\int (x - 1)e^{-x} dx = -(x - 1)e^{-x} - \int -e^{-x} dx = -e^{-x}(x - 1 + 1) = -x e^{-x} + C$$

$$u = (x - 1) \quad v = -e^{-x}$$

$$du = 1 dx \quad dv = -e^{-x} dx$$

$$y = \sqrt{v} \rightarrow y(x) = \sqrt{e^{-x} - x} \quad y(0) = 1 \rightarrow 1 = \sqrt{c} \rightarrow c = 1 \quad y(x) = \sqrt{e^{-x} - x}$$

Problema 6 (4 puntos)

Resuelve las siguientes ecuaciones diferenciales exactas:

a) $2x + y + xy \frac{dy}{dx} = 0$

b) $\frac{dy}{dx} = \frac{-2x - y}{x - 1}, \quad y(0) = 1$

c) $-\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy = 0, \quad y(1) = 2$

d) $x^2 + y^2 + 2y(x + 1) \frac{dy}{dx} = 0$

a) $2x + y + xy \frac{dy}{dx} = 0$

$(2x + y)dx + xy dy = 0 \quad \begin{cases} M = 2x + y \\ N = xy \end{cases}$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} ? \rightarrow \frac{\partial M}{\partial y} = 1 \quad \frac{\partial N}{\partial x} = y \quad \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \rightarrow \text{No es exacta}$

d) $x^2 + y^2 + 2y(x + 1) \frac{dy}{dx} = 0 \rightarrow \overbrace{(x^2 + y^2)}^M dx + \overbrace{2y(x + 1)}^N dy = 0$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} ? \rightarrow \frac{\partial M}{\partial y} = 2y \quad \frac{\partial N}{\partial x} = 2y \rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow \text{Es exacta}$

$\frac{\partial F}{\partial x} = x^2 + y^2 \rightarrow F(x, y) = \int x^2 + y^2 dx = \frac{x^3}{3} + xy^2 + g(y)$

$\frac{\partial F}{\partial y} = 2y(x + 1) \rightarrow 2yx + g'(y) = 2yx + 2y \rightarrow g'(y) = 2y \rightarrow g(y) = y^2 + C$

$\frac{x^3}{3} + xy^2 + y^2 = C \rightarrow C - \frac{x^3}{3} = xy^2 + y^2 \rightarrow y^2(x + 1) = C - \frac{x^3}{3} \rightarrow y^2 = \frac{C - \frac{x^3}{3}}{x + 1}$

$y = \pm \sqrt{\frac{C - \frac{x^3}{3}}{x + 1}}$

$$b) \frac{dy}{dx} = \frac{-2x-y}{x-1} \quad y(0) = 1$$

$$2x+y + (x-1) \frac{dy}{dx} = 0 \rightarrow (2x+y) dx + (x-1) dy = 0 \quad \begin{cases} M(x,y) = 2x+y \\ N(x,y) = (x-1) \end{cases}$$

$$\dot{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}? \quad \frac{\partial M}{\partial y} = 1 \quad \frac{\partial N}{\partial x} = 1 \rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow \text{Es exacta}$$

$$M = F_x = 2x+y \quad F(x,y) = \int 2x+y dx = x^2 + xy + g(y)$$

$$\frac{\partial F}{\partial y} = (x-1) \rightarrow x + g'(y) = (x-1) \rightarrow g'(y) = -1 \quad g(y) = -y + C$$

$$C = x^2 + xy - y \quad y(0) = 1 \rightarrow 0^2 + 0 \cdot 1 - 1 = C \rightarrow C = -1$$

$$F(x,y) = x^2 + xy - y + 1 \rightarrow x^2 + 1 = y - xy \rightarrow y(1-x) = x^2 + 1$$

$$y = \frac{x^2 + 1}{(1-x)} = \frac{-x^2 - 1}{x-1}$$

$$c) -\frac{y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy = 0 \quad y(1) = 2$$

$$M(x,y) = -\frac{y}{x^2+y^2} \quad N(x,y) = \frac{x}{x^2+y^2}$$

$$\dot{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}? \rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow \text{Es exacta}$$

$$\frac{\partial M}{\partial y} = -\frac{(x^2+y^2) - (y)(2y)}{(x^2+y^2)^2} = \frac{-x^2+y^2}{(x^2+y^2)^2} \quad \frac{\partial N}{\partial x} = \frac{(x^2+y^2) - x(2x)}{(x^2+y^2)^2} = \frac{-x^2+y^2}{(x^2+y^2)^2}$$

$$M = \frac{\partial F}{\partial x} = -\frac{y}{x^2+y^2} \rightarrow F(x,y) = \int \frac{-y}{x^2+y^2} dx = \int \frac{-y}{1+\frac{x^2}{y^2}} dx = -\arctg\left(\frac{x}{y}\right) + g(y)$$

$$\frac{\partial F}{\partial y} = \frac{2x}{x^2+y^2} + g'(y) = \frac{2x}{x^2+y^2} \rightarrow g'(y) = 0 \rightarrow g(y) = C$$

$$C = \arctg\left(\frac{x}{y}\right) \quad y(1) = 2 \rightarrow C = \arctg\left(\frac{1}{2}\right)$$

$$\arctg\left(\frac{1}{2}\right) = \arctg\left(\frac{x}{y}\right) \rightarrow \frac{x}{y} = \frac{1}{2} \rightarrow y = 2x$$